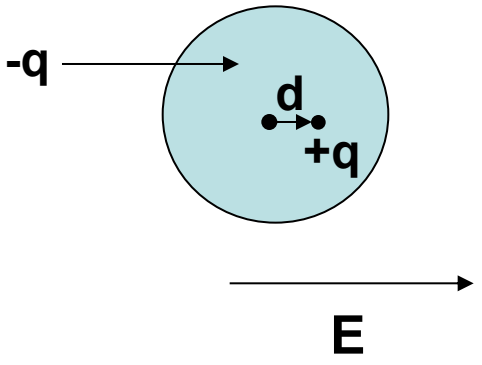
Polarization (don't confuse with polarization of light - This is electric polarization)

Dielectrics – what is it ?

What happens when a piece of dielectric is placed in an electric field

Induced Dipoles – what is it ?



Four types of polarizations

Atomic or electronic
Ionic
Orientational
Space Charge

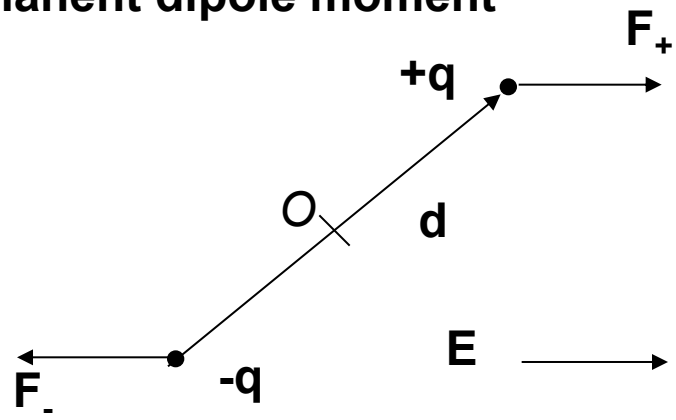
P = dipole moment per unit volume ***called polarization***

$$\vec{p} = \alpha \vec{E}$$

α is the atomic polarizability

Electric fields in matter

What happens when a molecule having a permanent dipole moment is kept in an electric field E



If the field is uniform, forces acting are

$$\vec{F}_+ = q\vec{E}$$

Cancels each other

$$\vec{F}_- = -q\vec{E}$$

Torque

$$\vec{N} = \left(\vec{r}_+ \times \vec{F}_+\right) + \left(\vec{r}_- \times \vec{F}_-\right)$$

$$= \left[\left(d/2\right) \times \left(q\vec{E}\right)\right] + \left[\left(-d/2\right) \times \left(-q\vec{E}\right)\right]$$

$$= q\vec{d} \times \vec{E} = \vec{p} \times \vec{E}$$

If the field is non-uniform, there will be net force in addition to the torque

$$\vec{F} = \vec{F}_+ + \vec{F}_- = q\left(\vec{E}_+ - \vec{E}_-\right) = q\left(\Delta\vec{E}\right)$$

$\Delta\vec{E}$ difference in field at the plus and minus ends

Since the dipole is very short, we can assume the small change in E_x as

$$\Delta E_x = \left(\nabla E_x\right)\cdot\vec{d}$$

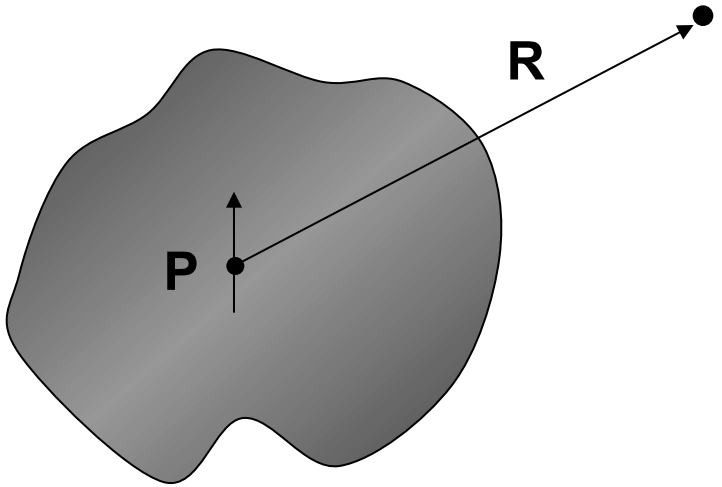
Together with corresponding formulas for E_y and E_z

$$\Delta\vec{E} = (\vec{d}\cdot\nabla)\vec{E}$$

$$\vec{F} = (\vec{p}\cdot\nabla)\vec{E}$$

The field due to a polarized object

Concept of Bound charges



Dipole moment per unit volume is **P**

For a single dipole **p** the potential

$$V(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{\hat{R}\cdot\vec{p}}{R^2}$$

For each volume element $\vec{p} = \vec{P}d\tau'$

Total potential is

$$V(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\int_V \frac{\hat{R}\cdot\vec{P}(\vec{r}')}{R^2}d\tau'$$

We know that $$\nabla'\left(\frac{1}{R}\right)=\frac{\hat{R}}{R^2}$$

Integrating with respect to this

$$V(\vec{r})=\frac{1}{4\pi\varepsilon_0}\int_v \vec{P}\cdot\nabla'\left(\frac{1}{R}\right)d\tau'$$

Using product rule $$\vec{\nabla}.(\phi\vec{E})=\vec{\nabla}\phi.\vec{E}+\phi\vec{\nabla}.\vec{E}$$

$$V=\frac{1}{4\pi\varepsilon_0}\left[\int_v \nabla'\cdot\left(\frac{\vec{P}}{R}\right)d\tau' - \int_v \frac{1}{R}\left(\nabla'\cdot\vec{P}\right)d\tau'\right]$$

Use divergence theorem

$$V=\frac{1}{4\pi\varepsilon_0}\oint_S \frac{1}{R}\vec{P}\cdot d\vec{a}' - \frac{1}{4\pi\varepsilon_0}\int_v \frac{1}{R}\left(\nabla'\cdot\vec{P}\right)d\tau'$$

$$V = \frac{1}{4\pi\varepsilon_0}\oint_S \frac{1}{R}\vec{P}\cdot d\vec{a}' - \frac{1}{4\pi\varepsilon_0}\int_v \frac{1}{R}\left(\nabla'\cdot\vec{P}\right)d\tau'$$

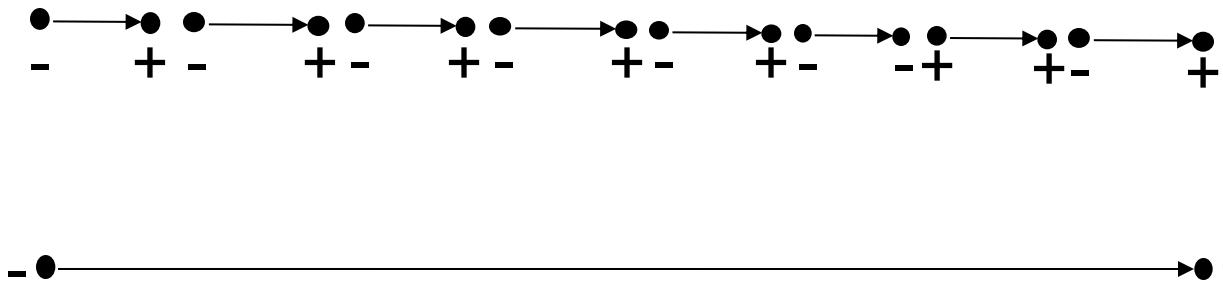
First term is potential produced by a surface charge density σ_b and second term is that produced by a volume charge density ρ_b

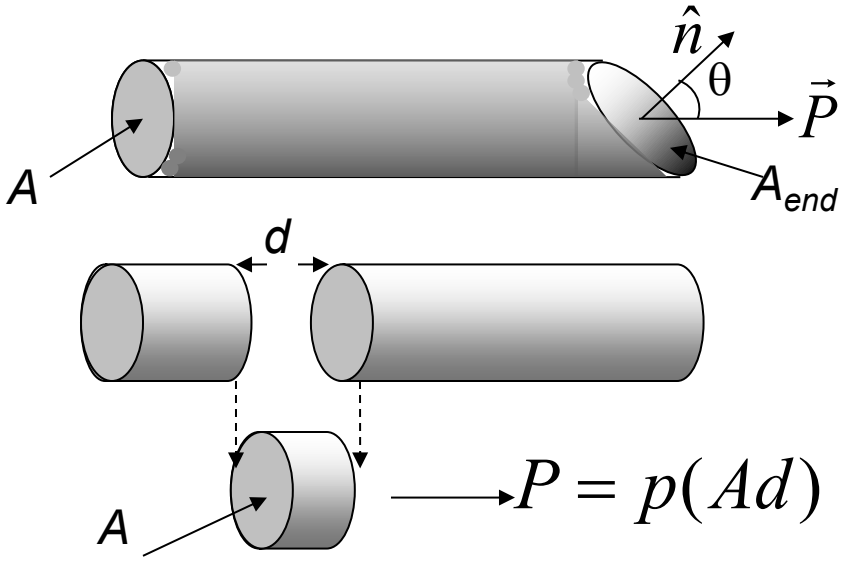
$$\sigma_b = \vec{P}\cdot\hat{n} \qquad \rho_b = -\vec{\nabla}\cdot\vec{P}$$

$$V = \frac{1}{4\pi\varepsilon_0}\oint_S \frac{\sigma_b}{R}d\vec{a}' + \frac{1}{4\pi\varepsilon_0}\int_v \frac{\rho_b}{R}d\tau'$$

Potential of a polarized object is the same as that produced by a volume charge density ρ_b plus that produced by a surface charge density σ_b
Instead of integrating infinitesmal dipoles, we just find the field due to those bound charges.

Physical Interpretation of Bound Charges





p – dipole moment
P - polarization

In terms of charge q at the ends, the dipole moment is written as

$$p = qd \Rightarrow q = PA$$

$$\sigma_b = \frac{q}{A} = P$$ If the end is sliced off perpendicularly

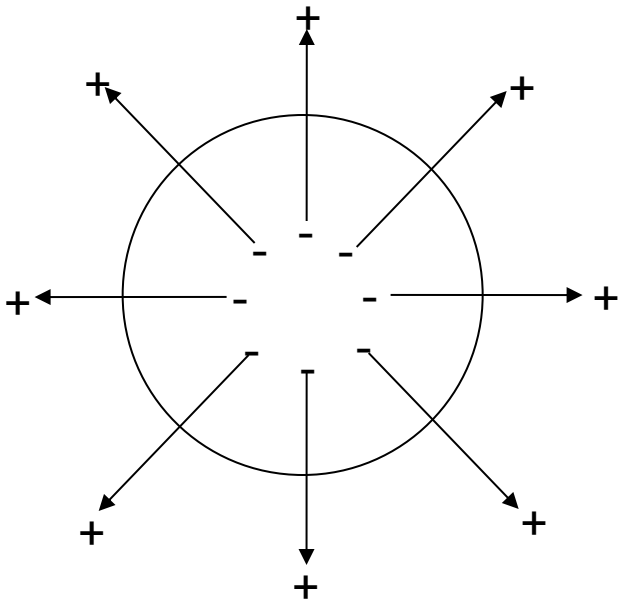
$$A = A_{end} \cos\theta$$ If the end is sliced off at an angle

$$\therefore \sigma_b = \frac{q}{A_{end}} = P\cos\theta = \vec{P} \cdot \hat{n}$$

The effect of polarization is to paint a bound charge σ_b over the surface

The net bound charge $\int \rho_b d\tau$ in a given volume is equal and opposite to the amount of charge that has been pushed out through the surface, if the polarization is non-uniform.

$$\int_V \rho_b d\tau = -\oint_s \vec{P} \cdot \overrightarrow{da} = -\int_V \left(\nabla \cdot \vec{P}\right) d\tau$$



Since this is true for any volume $\Rightarrow \rho_b = -\nabla \cdot \vec{P}$

The Electric Displacement

Gauss's law in the presence of dielectrics

Effect of polarization is to produce accumulation of bound charges

$\sigma_b = \vec{P} \cdot \hat{n}$ on the surface, $\rho_b = -\vec{\nabla} \cdot \vec{P}$ within dielectric

Free charge: any charge that is not a result of polarization ρ_f

Field attributed to bound charge and that due to everything else within the dielectric is the total charge density given by

$$\rho = \rho_b + \rho_f$$

Gauss's law states that

$$\varepsilon_0 \vec{\nabla} \cdot \vec{E} = \rho = \rho_b + \rho_f = -\vec{\nabla}.\vec{P} + \rho_f$$

$\vec{E}$ Is now the total field, not *just* that portion generated by polarization

$$\varepsilon_0 \vec{\nabla} \cdot \vec{E} = -\vec{\nabla}.\vec{P} + \rho_f$$

$$\vec{\nabla}.(\varepsilon_0 \vec{E} + \vec{P}) = \rho_f$$

Term in the bracket is replaced by $\vec{D}$

$$\vec{D} = \varepsilon_0 \vec{E} + \vec{P}$$

is the electric displacement

$$\vec{\nabla} \cdot \vec{D} = \rho_f$$

In the integral form

$$\oint \vec{D} \cdot d\vec{a} = Q_{f_{enc}}$$

$Q_{f_{enc}}$ is the total free charge enclosed in the volume

Linear Dielectrics

For many substances polarization is proportional to the field provided the field is not too strong

$$\vec{P} = \varepsilon_0 \chi_e \vec{E}$$

χ_e Is the dielectric susceptibility

Materials that obey this equation are the linear dielectrics

In linear media

$$\vec{D} = \varepsilon_0 \vec{E} + \vec{P} = \varepsilon_0 \vec{E} + \varepsilon_0 \chi_e \vec{E} = \varepsilon_0 (1 + \chi_e) \vec{E}$$

So $\vec{D}$ is also proportional to $\vec{E}$

$$\vec{D} = \varepsilon \vec{E} \qquad \varepsilon = \varepsilon_0 (1 + \chi_e)$$ is the permittivity of the materials

$$\varepsilon = \varepsilon_0 (1 + \chi_e) \qquad \varepsilon_r = \frac{\varepsilon}{\varepsilon_0} = (1 + \chi_e)$$

is the relative permittivity or dielectric constant of the material

Energy in Dielectric systems

We have seen $W = \frac{\varepsilon_0}{2} \int_{\text{all space}} |\vec{E}|^2 \, dV$ $W = \frac{1}{2} \int \phi \rho dV$

Work done on the incremental *free* charge

$$\Delta W = \frac{1}{2} \int (\Delta \rho_f) V d\tau$$

$$\vec{\nabla} \cdot \vec{D} = \rho_f \qquad \Delta \rho_f = \vec{\nabla} \cdot (\Delta \vec{D})$$

$$\Delta W = \frac{1}{2} \int (\vec{\nabla} \cdot (\Delta \vec{D})) V d\tau$$

Using product rule $\vec{\nabla}.(\phi \vec{E}) = \vec{\nabla}\phi.\vec{E} + \phi \vec{\nabla}.\vec{E}$

$$\vec{\nabla} \cdot \left[(\Delta \vec{D}) V \right] = \left[\vec{\nabla} \cdot (\Delta \vec{D}) V \right] + \Delta \vec{D} \cdot (\vec{\nabla} V)$$

$$\Delta W = \frac{1}{2} \int \vec{\nabla} \cdot \left[(\Delta \vec{D}) V \right] d\tau + \frac{1}{2} \int (\Delta \vec{D}) \cdot \vec{E} d\tau$$

The divergence theorem turns the first integral into a surface integral, which vanishes if we integrate over all space

$$\Delta W = \frac{1}{2}\int \left(\Delta \vec{D}\right)\cdot \vec{E} d\tau$$

If the medium is a linear dielectric $\vec{D} = \varepsilon \vec{E}$

so $$\frac{1}{2}\Delta\left(\vec{D}\cdot\vec{E}\right) = \frac{1}{2}\Delta\left(\varepsilon E^2\right) = \frac{1}{2}\varepsilon\left(\Delta\vec{E}\right)\cdot\vec{E} = \frac{1}{2}\left(\Delta\vec{D}\right)\cdot\vec{E}$$

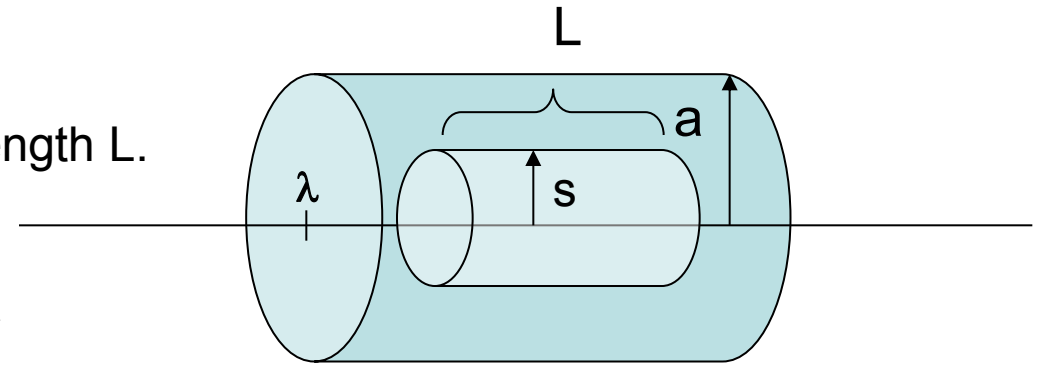
Thus $$\Delta W = \Delta\left(\frac{1}{2}\int \vec{D}\cdot\vec{E} d\tau\right)$$

Hence the total work done

$$W = \frac{1}{2}\int \vec{D}\cdot\vec{E} d\tau$$

A long straight line carrying uniform line charge λ, is surrounded by rubber insulation out to a radius *a.* Find the electric displacement

Gaussian surface of radius s and length L.



$$D(2\pi sL) = \lambda L$$

$$\vec{D} = \frac{\lambda}{2\pi s}\hat{s}$$

This equation holds good for both inside and outside the rubber insulation

Outside **P** = 0 $\vec{E} = \frac{1}{\varepsilon_0}\vec{D} = \frac{\lambda}{2\pi\varepsilon_0 s}\hat{s}$ for s > a

Inside the rubber electric field cannot be determined, since we do not know **P**

$\vec{D}$ and $\vec{E}$

There is no coulomb's law for D

Curl $\vec{E}$ is always zero, but Curl $\vec{D}$ is not

$$\vec{\nabla} \times \vec{D} = \vec{\nabla} \times \left(\varepsilon_0 \vec{E} + \vec{P} \right) = \vec{\nabla} \times \vec{P}$$

It will depend on **P**

Boundary conditions

$$D^{\perp}_{above} - D^{\perp}_{below} = \sigma_f$$

$$D^{||}_{above} - D^{||}_{below} = P^{||}_{above} - P^{||}_{below}$$

Corresponding boundary conditions for E are

$$E^{\perp}_{above} - E^{\perp}_{below} = \frac{\sigma}{\varepsilon_0}$$

$$E^{||}_{above} - E^{||}_{below} = 0$$

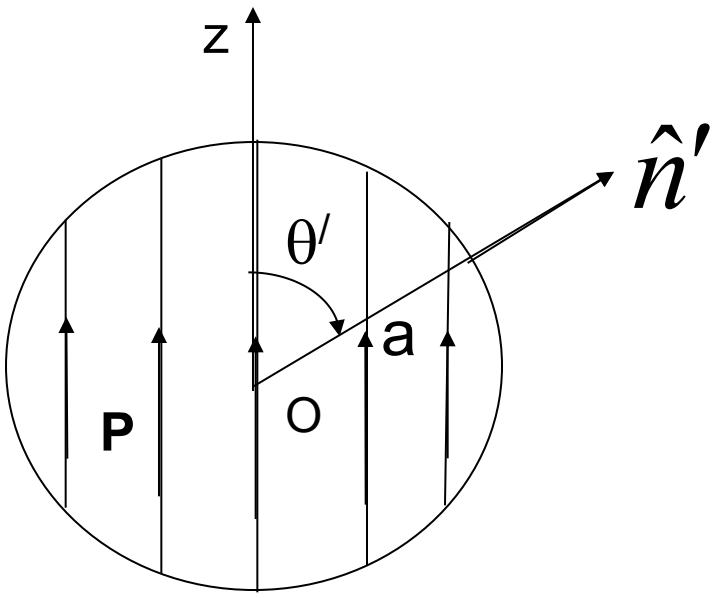
Uniformly polarized sphere

A sphere of radius ***a*** has constant polarization **P**

Choose z axis to be the direction of P

Origin at the center of the sphere

$$\vec{P} = P\hat{z}$$



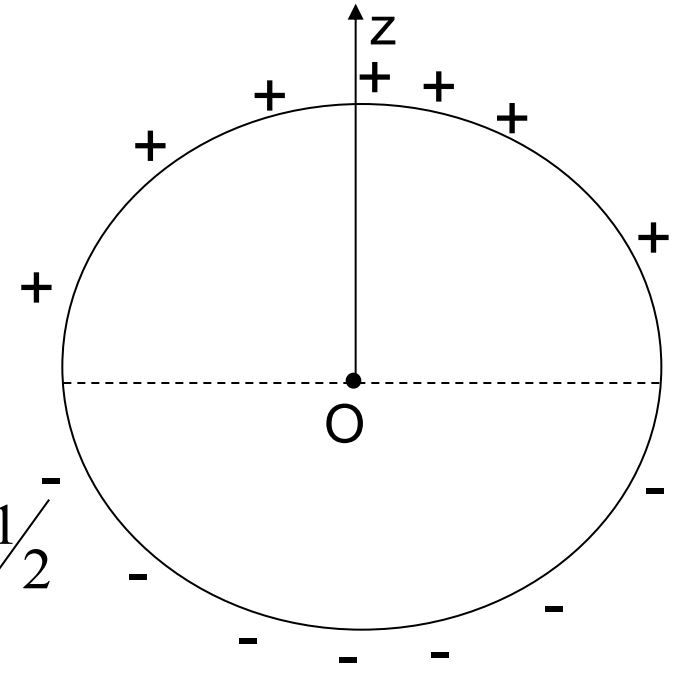
P is a constant and hence volume bound charge is zero

But have a finite surface bound charge

$$\sigma_b(\theta') = \vec{P}.\hat{n}' = P\cos\theta'$$

$$\sigma_b(\theta') = \vec{P}.\hat{n}' = P\cos\theta'$$

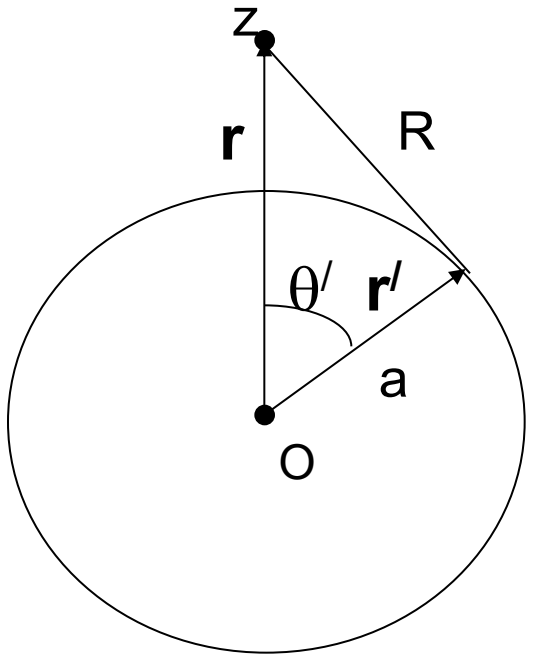
For simplicity we find the potential and field on points on the z axis and for positive z



$$R = \left(z^2 + a^2 - 2za\cos\theta'\right)^{1/2}$$

Potential is

$$\phi(z) = \frac{1}{4\pi\varepsilon_0}\int \frac{\sigma_b(\theta')da'}{R}$$



$$\phi(z) = \frac{1}{4\pi\varepsilon_0}\int \frac{\sigma_b(\theta')da'}{R}$$

$$= \frac{1}{4\pi\varepsilon_0}\int_0^{2\pi}\int_0^{\pi} \frac{P\cos\theta' \cdot a^2 \sin\theta' d\theta' d\phi'}{\left(z^2 + a^2 - 2za\cos\theta'\right)^{1/2}}$$

$$= \frac{Pa^2}{2\varepsilon_0}\int_{-1}^{1} \frac{\mu d\mu}{\left(z^2 + a^2 - 2za\mu\right)^{1/2}}$$

This turns out to be

$$-\frac{\left(z^2 + a^2 + za\mu\right)\left(z^2 + a^2 - 2za\mu\right)^{1/2}}{3z^2a^2}\Bigg|_{-1}^{1}$$

$$\int \frac{\mu d\mu}{(z^2+a^2-2za\mu)^{1/2}}$$

$$z^2+a^2-2za\mu = \rho^2$$

$$\Rightarrow -2za\, d\mu = 2\rho d\rho$$

$$\mu = \frac{z^2+a^2-\rho^2}{2za}$$

$$\int -\frac{(z^2+a^2-\rho^2)}{2za} \frac{-\rho d\rho}{za\cdot\rho}$$

$$= \frac{-1}{2z^2a^2}\left[(z^2+a^2)\rho - \rho^3/3\right] =$$

$$-\frac{\left(z^2+a^2+za\mu\right)\left(z^2+a^2-2za\mu\right)^{1/2}}{3z^2a^2}\Bigg|_{-1}^{1}$$

$$=\frac{1}{3z^2a^2}\left[\left(z^2+a^2\right)\left(|z+a|-|z-a|\right)-za\left(|z+a|+|z-a|\right)\right]$$

(1)

We have to discuss two situations

1.Outside the sphere

Here z > a $\quad |z-a| = z-a$

(1) Becomes $2a / 3z^2$

$$\phi_o(z)=\frac{Pa^3}{3\varepsilon_0 z^2}$$

Electric field

$$E_{zo}(z) = -\frac{\partial \phi_0}{\partial z} = \frac{2Pa^3}{3\varepsilon_0 z^3}$$

We can express the same in terms of the total dipole moment of the sphere

$$\vec{p} = \frac{4}{3}\pi a^3 P\hat{z}$$

$$\phi_o(z) = \frac{p}{4\pi\varepsilon_0 z^2}$$

$$E_{zo}(z) = \frac{2p}{4\pi\varepsilon_0 z^3}$$

Inside the sphere

Here z < a, so that $|z-a| = a - z$

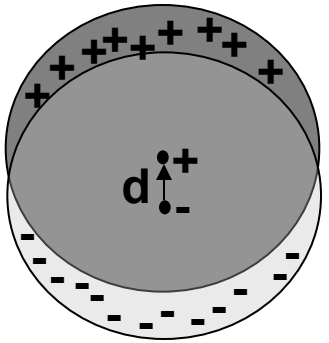
(1) becomes $2z/3a^2$

Potential becomes $$\phi_i(z) = \frac{Pz}{3\varepsilon_0}$$

and field $$E_{zi}(z) = -\frac{\partial \phi_i}{\partial z} = -\frac{P}{3\varepsilon_0}$$

E is a constant inside the sphere

Field in the region of overlap between two spheres



$$\vec{E} = -\frac{1}{4\pi\varepsilon_0}\frac{q\vec{d}}{R^3}$$

$$\vec{p} = q\vec{d} = \left(\frac{4}{3}\pi R^3\right)\vec{P}$$

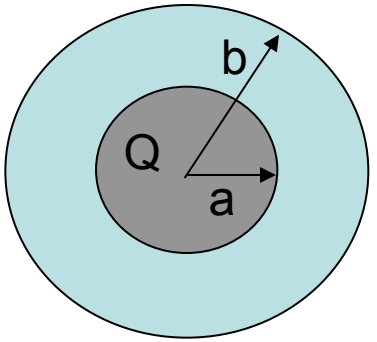
$$\vec{E} = -\frac{1}{3\varepsilon_0}\vec{P}$$

A metal sphere of radius a carries a charge Q. It is surrounded, out to radius b, by linear dielectric material of permittivity ε. Find the potential at the center (relative to infinity)

Free charge here is Q

The arrangement if spherically symmetric.

$$\vec{D} = \frac{Q}{4\pi r^2}\hat{e}_r$$ for r > a



Inside the metal sphere, **E** = **P** = **D** = 0

Once **D** is known, it is easy to find **E**

$$\vec{E} = \begin{cases} \dfrac{Q}{4\pi\varepsilon r^2}\hat{e}_r & \text{for a < r < b} \\ \dfrac{Q}{4\pi\varepsilon_0 r^2}\hat{e}_r & \text{for r > b} \end{cases}$$

Potential at the center is

$$\vec{V} = -\int_{\infty}^{0} \vec{E}\cdot d\vec{l} = -\int_{\infty}^{b}\left(\frac{Q}{4\pi\varepsilon_0 r^2}\right)dr - \int_{b}^{a}\left(\frac{Q}{4\pi\varepsilon r^2}\right)dr - \int_{a}^{0}(0)dr$$

$$= \frac{Q}{4\pi}\left(\frac{1}{\varepsilon_0 b} + \frac{1}{\varepsilon a} - \frac{1}{\varepsilon b}\right)$$

$$\vec{P} = \varepsilon_0\chi_0\vec{E} = \frac{\varepsilon_0\chi_0 Q}{4\pi\varepsilon r^2}\hat{e}_r$$

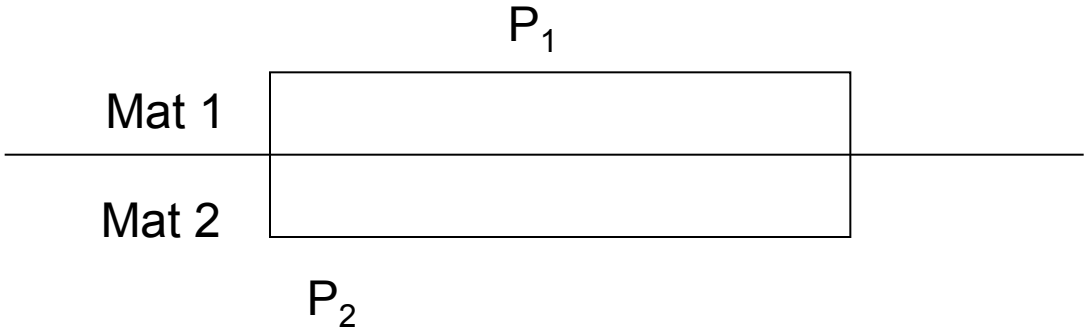
Volume bound charge $\rho_b = 0$

$$\rho_b = -\vec{\nabla} \cdot \vec{P} = 0$$

$$\sigma_b = \vec{P} \cdot \hat{n} = \begin{cases} \dfrac{\varepsilon_0 \chi_0 Q}{4\pi \varepsilon b^2} & \text{at the outer surface} \\ \dfrac{-\varepsilon_0 \chi_0 Q}{4\pi \varepsilon a^2} & \text{at the inner surface} \end{cases}$$

Note the – sign at the inner surface

Can curl of D vanish if we are dealing with linear dielectrics?





If the space is filled with a homogeneous linear dielectric, then

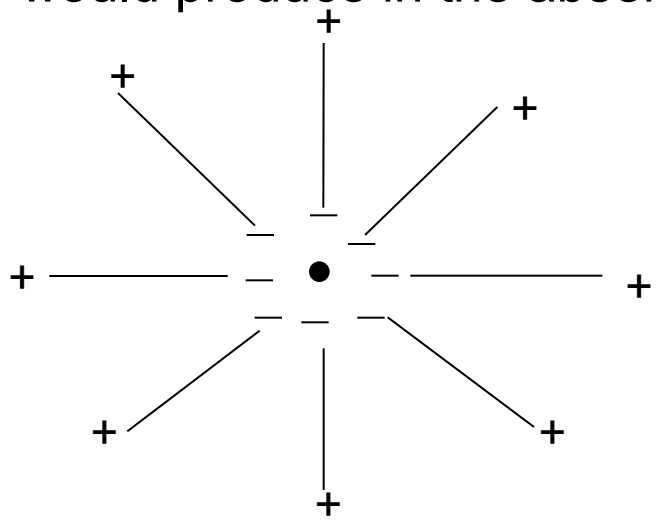
$$\vec{\nabla} \cdot \vec{D} = \rho_f \qquad \vec{\nabla} \times \vec{D} = 0$$

In this case D can be found from the free charge as though the dielectric was not there

$$\vec{D} = \varepsilon_0 \vec{E}_{vac}$$

$\vec{E}_{vac}$ is the field the same free charge would produce in the absence of any dielectric

$$\vec{E} = \frac{1}{\varepsilon}\vec{D} = \frac{1}{\varepsilon_r}\vec{E}_{vac}$$



In a homogeneous linear dielectric the bound charge density is proportional to the free charge density

$$\rho_b = -\vec{\nabla}\cdot\vec{P} = -\vec{\nabla}\cdot\left(\varepsilon_0\frac{\chi_e}{\varepsilon}\vec{D}\right) = -\left(\frac{\chi_e}{1+\chi_e}\right)\rho_f$$

Unless free charge is embedded in the material, $\rho = 0$
Within such a dielectric the potential obeys Laplace's equation